
Class-Conditional Distribution Balancing for Group Robust Classification

Miaoyun Zhao^{1,2} Qiang Zhang²

Abstract

Spurious correlations that lead models to correct predictions for the wrong reasons pose a critical challenge for robust real-world generalization. Existing research attributes this issue to group imbalance and addresses it by maximizing group-balanced or worst-group accuracy, which heavily relies on expensive bias annotations. A compromise approach involves predicting bias information using extensively pretrained foundation models, which requires large-scale data and is limited to physically interpretable biases. To address these challenges, we offer a novel perspective by reframing the spurious correlations as imbalances/mismatches in class-conditional distributions caused by general biases, whether interpretable or not, and propose a simple yet effective robust learning method that mitigates the need for bias annotations or predictions. With the goal of maximizing the conditional entropy (uncertainty) of the label given spurious factors, our method leverages a sample reweighting strategy to achieve class-conditional distribution balancing, which automatically highlights minority groups and classes, effectively mitigating spurious correlations and producing a debiased data distribution for classification. Extensive experiments and analysis demonstrate that our approach consistently delivers strong overall performance, and is competitive with methods that rely on bias supervision.

1. Introduction

Despite the unprecedented success of deep learning in recent decades, machine learning methods following the ERM

paradigm (Vapnik, 1999) often suffer from spurious correlations when training data is biased: some spurious factors (also termed bias) are predictive of the target class solely due to their high co-occurrence with the target label, despite being irrelevant to the true classification function (Geirhos et al., 2020; LaBonte et al., 2023). It causes unpredictable performance losses under distribution shift, where spurious correlations in the test data differ from or conflict with those in the training data (Shah et al., 2020; Zhang et al., 2021). For example, in the task of recognizing cows and camels (Tsirigotis et al., 2023), the training dataset is biased with camel images predominantly associating with desert backgrounds, leaving camels on grass as a minority group (Beery et al., 2018). Consequently, the classifier is likely to associate camels with deserts, resulting in a biased model that fails the underrepresented groups (Tatman, 2017). Beyond these simple cases, spurious correlations pose a major challenge in many high-stakes settings where fairness is critical (Caton & Haas, 2024).

While progress has been made, existing research attributes spurious correlations to group imbalance. They enhance robustness against spurious correlations by utilizing bias information to implement group balanced measures during training (Bahng et al., 2020; Sagawa et al., 2020; Kim et al., 2024) or validation (Nam et al., 2020; Liu et al., 2021; 2023; Kirichenko et al., 2022; Zhou et al., 2022). These methods can be categorized into bias-supervised and pseudo-bias-supervised methods, depending on how bias information is obtained. Bias-supervised methods (Bahng et al., 2020; Sagawa et al.; Liu et al., 2021; 2023; Kirichenko et al., 2022; Zhou et al., 2022) rely on ground-truth bias annotations provided by human annotators, which limits their applicability in practical scenarios where such annotations are too costly or even impossible to obtain. To address these limitations, pseudo-bias-supervised methods (Asgari Taghanaki et al., 2022; LaBonte et al., 2023; Tsirigotis et al., 2023; Pezeshki et al., 2024; Kim et al., 2024) are developed to use predicted bias information instead of human annotations. By leveraging extensively pretrained large scale visual language models (Mokady et al., 2021; Li et al., 2022b), some methods achieve performance only slightly inferior to that of bias-supervised approaches. However, they mainly target physically interpretable biases and may not generalize to more abstract biases that are not predefined (e.g., noise in

¹Ningbo Institute of Dalian University of Technology, Ningbo 315016, China. Email: miaoyun9zhao@gmail.com. ²Key Laboratory of Social Computing and Cognitive Intelligence, Dalian University of Technology, Dalian, China. Correspondence to: Qiang Zhang <zhangq@dlut.edu.cn>.

cCIFAR10 (Hendrycks & Dietterich, 2018)).

Here we consider a more realistic and challenging scenario where neither bias annotations nor a pretrained model for bias prediction are available. With this goal, a line of research on stable learning has been proposed (Kuang et al., 2020; Zhang et al., 2021; Yu et al., 2023). These methods assume a structural decomposition of features into core and spurious factors and attribute failures under distribution shift to their statistical dependence. They propose global sample reweighting to mitigate spurious correlations by decorrelating features without explicit bias information. However, with finite samples, perfect independence is theoretically hard to achieve. Moreover, most methods are designed for linear models with predefined features; when extended to deep models, their effectiveness degrades due to feature entanglement, where decorrelation alone fails to ensure proper disentanglement (Yu et al., 2023).

Motivated by these observations, we explore an orthogonal approach that maximizes the conditional entropy of the label given spurious factors through sample reweighting, enabling the model to make decisions free from spurious correlations. We offer a novel perspective on the origins of spurious correlations: (i) distribution imbalance, referring to the class-conditional distribution mismatch, and (ii) class imbalance. Accordingly, we design a modified class-specific reweighting mechanism to address both types of imbalance. Building on this, we propose the class-conditional distribution balancing (CCDB) technique, which automatically reduces label predictability arising from spurious factors and thus yields a data distribution with spurious correlations removed. We further provide an analysis suggesting that CCDB generalizes traditional class-balancing methods by incorporating conditional distribution matching, and that this matching is consistent with the “covariate balance” principle in causal inference (Neal, 2020).

The main contributions of this work are as follows:

- (i) We design a class-specific sample reweighting mechanism and, based on it, a class-conditional distribution balancing technique that generalizes class balancing by adjusting the data from both class-conditional and class-marginal distributions;
- (ii) By maximizing the entropy of the target label given spurious factors, we design a three-stage procedure to produce a reweighted data distribution that mitigates spurious correlations;
- (iii) Our method saves the labor-intensive work of bias annotations or predictions. Extensive experiments on various synthetic and real-world datasets demonstrate its superiority over methods that rely on bias supervision.

2. Related work

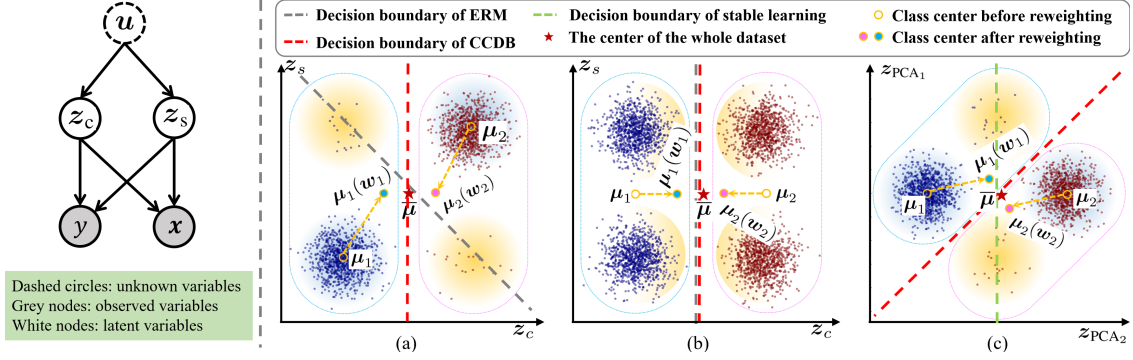
2.1. Bias-guided methods

Addressing spurious correlations becomes straightforward with bias annotations. By combining both bias and label annotations, one can create a group-balanced training process (e.g., GroupDRO (Sagawa et al., 2020)) or model selection strategy (e.g., DFR (Kirichenko et al., 2022), MAPLE (Zhou et al., 2022)) to enhance robustness against spurious correlations. To reduce reliance on manual annotations, pseudo-bias-supervised methods propose using predicted bias information instead of explicit annotations, typically via auxiliary models, and focusing on data from the worst-performing groups (e.g., LFF (Nam et al., 2020), LC (Liu et al., 2023), DFA (Chu et al., 2021), JTT (Liu et al., 2021)). Additionally, recent research (Kim et al., 2024; Yu et al., 2024) explores integrating large-scale vision-language models for more accurate bias prediction. However, their performance depends on prediction quality and still requires bias annotations for optimal performance.

2.2. Bias-agnostic methods

To mitigate the dependence on bias annotations/predictions, researchers have developed bias-agnostic methods driven by different motivations. One line of work focuses on stable learning (Kuang et al., 2018), which attributes model failure under distribution shifts to the statistical dependency between core and spurious features. Early approaches (Kuang et al., 2020; Zhang et al., 2021) aim to mitigate spurious correlations by decorrelating feature dimensions. These methods are primarily developed under linear frameworks with predefined features, making their extension to deep models challenging. Follow-up works (Lin et al., 2022; Yu et al., 2023) extend stable learning to deep models through invariance learning or sparse independence regularization without explicit environment partition. Nevertheless, learning stable representations in deep models remains challenging due to the heavy entanglement between core and spurious factors, making invariance or decorrelation constraints alone insufficient for reliably separating them. Another line of research utilizes disagreements among diverse models to improve generalization under distribution shifts (Lee et al., 2022; Pagliardini et al., 2023; LaBonte et al., 2023), but these methods still require a small set of group annotations for model selection. Several other bias-agnostic methods have also demonstrated competitive performance (Asgari Taghanaki et al., 2022; Han et al., 2023; Tsirigotis et al., 2023; Pezeshki et al., 2024; Asaad et al., 2025). However, most of these methods have difficulty balancing minority- and majority-group performance, thereby requiring careful hyperparameter tuning, often based on bias-annotated validation data.

Our method also falls within the bias-agnostic framework.



Unlike the aforementioned methods, it (i) avoids decorrelating all features and instead maximizes the conditional entropy of labels given spurious factors, and (ii) enables simple control of sample weighting via distribution distance minimization. Note, (Moyer et al., 2018) also studies mutual information for learning representations invariant to given attributes. In contrast, our method does not rely on explicit MI minimization, but instead maximizes conditional entropy, where MI arises implicitly. Please refer to the appendix for a detailed discussion of related work.

3. Our method

Given each observation consists of an input x (e.g., an image $x \in \mathbb{R}^{L \times H \times W}$) and a label $y \in \mathbb{R}$, where L , H , and W represent the channel, height and width of the image, respectively, the spurious correlation problem is defined as follows. Let $z \in \mathbb{R}^D$ denote the latent representation of x after feature extraction, which contains core and spurious factors, $z = [z_c, z_s]$. Here, $z_c \in \mathbb{R}^{D_c}$ and $z_s \in \mathbb{R}^{D_s}$ are implicitly entangled and technically challenging to decorrelate. D , D_c , and D_s denote feature dimensions. Ideally, z_c captures label-relevant information and is strongly correlated with y , while z_s is unrelated to the label. However, under distribution shift, the presence of unknown confounders (Pearl, 2009) induces strong spurious correlations between z_s and z_c , and consequently between z_s and y in training data (Christiansen et al., 2021). This makes z_s highly predictive of y during training, but unreliable at test time, undermining model generalization. To address this issue, we break spurious correlations by reshaping the joint distribution $p(z_s, y)$. Specifically, we make z_s non-predictive of y by maximizing the conditional uncertainty of y given z_s , i.e., minimizing the negative conditional entropy as follows,

$$\begin{aligned} \mathcal{L} &= -H(y|z_s) = I(z_s, y) - H(y) \\ &= \mathbb{E}_{p(y)} \text{KL}[p(z_s|y) \| p(z_s)] + \mathbb{E}_{p(y)} \log p(y) \end{aligned} \quad (1)$$

where $H(\cdot)$ and $I(\cdot)$ denote entropy and mutual information, and $\text{KL}[\cdot \| \cdot]$ denotes the Kullback–Leibler divergence

(Kullback & Leibler, 1951). Intuitively, Equation (1) shows that spurious correlations arise from two primary factors: (i) the mismatch among the class-conditional distributions, termed distribution imbalance and (ii) the non-uniformity of the class distribution, termed class imbalance. Motivated by these key insights, we propose the Class-conditional distribution balancing (CCDB) technique to break spurious correlations by minimizing Equation (1), which involves reducing the mutual information between z_s and y while maximizing the entropy of y . Note that the first term in Equation (1) drives the class-conditional distributions of z_s toward a common marginal distribution, which can be interpreted as distribution-level balancing and is reminiscent of covariate balance in causal inference (Neal, 2020; Yin et al., 2024). Given the causal graph in Figure 1 (left), taking the binary classification as an example, treating z_c as the treatment and y as the outcome induces a causal path $z_c \rightarrow y$ as well as a backdoor path $z_c \leftarrow u \rightarrow z_s \rightarrow y$. The spurious correlation between z_s and y arises from the unknown confounder u . To safely estimate the causal effect of an image feature z on the label y , one must remove the confounding bias induced by the differing distributions of z_s between the treated and control image sets¹. This encourages $p(z_s|z_c) = p(z_s)$, which is consistent with the goal of achieving $p(z_s|y) = p(z_s)$ in the KL term, under the assumption that y is strongly correlated with z_c . The second term in Equation (1) promotes uniformity across classes, ensuring class balancing. Thus, Equation (1) generalizes traditional class-balancing methods by addressing both distributional and class-level imbalances within a unified framework. Note, Equation (1) identifies spurious features via mismatches between class-conditional and marginal distributions, enabling CCDB to handle unknown and complex biases as long as they induce such mismatches.

Directly minimizing Equation (1) is nontrivial because the disentangled z_s and its exact distributions are not accessible.

¹Since u is unknown, using z_s provides an equivalent way to block the backdoor path.

For practical implementation, we first propose learning a biased feature extractor in Section 3.2, which captures features dominated by spurious correlations, *i.e.*, $z \approx z_s$, and use these features to conduct spurious correlation mitigation. **Accordingly, to avoid cumbersome notation, we denote z_s simply as z in the following sections.** Secondly, we introduce the following designations to handle each term in Equation (1) separately.

The KL term. By approximating the data distribution with a Gaussian², we adopt the 2-Wasserstein distance (Mallasto & Feragen, 2017; Arjovsky et al., 2017) as a substitute to measure the discrepancy between $p(z|y)$ and $p(z)$:

$$W_2(z|_{y=k}, z) = \|\mu_k - \bar{\mu}\|_2^2 + d(C_k, \bar{C}) \quad (2)$$

where, $d(C_k, \bar{C}) = \text{trace}(C_k + \bar{C} - 2[\bar{C}^{\frac{1}{2}} C_k \bar{C}^{\frac{1}{2}}]^{\frac{1}{2}})$, $\{\mu_k, C_k\}$ and $\{\bar{\mu}, \bar{C}\}$ denote the mean and covariance matrix for $p(z|y=k)$ and $p(z)$, respectively.

The entropy term. This term can be easily maximized through data subsampling or reweighting, following traditional class-balancing techniques (Idrissi et al., 2022). In fact, it is possible to handle these two terms together. In the following section, we propose a modified sample reweighting mechanism to achieve this goal.

3.1. Intra-class sample reweighting for CCDB

Inspired by the success of sample reweighting for decorrelation (Zhang et al., 2021; Yu et al., 2023), we leverage it to mitigate spurious correlations between z and y . Unlike prior methods that learn a reweighting scheme over the entire training data, we apply reweighting independently within each class, enabling automatic optimization of both terms in Equation (1).

Suppose the observations come from K classes, with class k represented as $\{x_i, y_i = k\}_{i=1}^{n_k}$, where n_k is the number of samples in class k . We introduce class-specific sample weights $w_k \in \mathbb{R}^{n_k}$ to adjust the class-conditional distribution, subject to the class-wise simplex constraints: $[w_k]_i \geq 0$ for $i = 1, \dots, n_k$, and $\sum_{i=1}^{n_k} [w_k]_i = 1$. After reweighting, the distribution distance between $p(z|y=k, w_k)$ and $p(z)$ following Equation (2) is computed as:

$$W_2(z|_{\{y=k, w_k\}}, z) = \|\mu_k(w_k) - \bar{\mu}\|_2^2 + d(C_k(w_k), \bar{C}) \quad (3)$$

where $\mu_k(w_k)$ and $C_k(w_k)$ denote the mean and covariance of class k after reweighting, defined as,

$$\begin{aligned} \mu_k(w_k) &= \sum_{i=1, y_i=k}^{n_k} [w_k]_i \cdot z_i \\ C_k(w_k) &= \sum_{i=1, y_i=k}^{n_k} [w_k]_i (z_i - \mu_k(w_k))(z_i - \mu_k(w_k))^\top \end{aligned} \quad (4)$$

Note, we set $\bar{\mu}$ and $\bar{C}$ as constants rather than functions of $\{w_1, \dots, w_K\}$ for two reasons: (i) The marginal distribution $(\bar{\mu}, \bar{C})$ acts as a stable anchor for the class-conditional distributions; ideally, if all class-conditionals match the marginal, it remains unchanged. (ii) making $\bar{\mu}$ and $\bar{C}$ dependent on $\{w_1, \dots, w_K\}$, would make the objective overly flexible, potentially leading to degenerate solutions with many possible marginal distributions. Accordingly, the objective in Equation (1) is reformulated as follows,

$$\mathcal{L}(w_1, \dots, w_K) = \frac{1}{K} \sum_{k=1}^K W_2(z|_{\{y=k, w_k\}}, z) \quad (5)$$

where we omit the entropy term because w_k , being class-specific and normalized to sum to 1, naturally assigns larger weights to underrepresented classes. Together with the “*WeightedRandomSampler*” used in Stage 3 of Section 3.2, this effectively mitigates the long-tailed distribution problem.

Derivation of CCDB for Long-Tailed Distribution Mitigation. Let the original empirical dataset contains N samples from K classes, with distribution $p_0(x, y)$, and let $a = [w_1, \dots, w_K]$ denote the concatenation of the learned sample weights. The class-wise simplex constraint enforces normalization within each class, *i.e.*, for class k , $\sum_{i=1}^N a_i \mathbf{1}_{y_i=k} = 1$, where $\mathbf{1}_{y_i=k}$ is the indicator function that equals 1 when $y_i = k$, and 0 otherwise. The *WeightedRandomSampler* draws each sample (x_i, y_i) with probability:

$$p_1(x_i, y_i) = \frac{a_i}{\sum_{j=1}^N a_j} = \frac{a_i}{\sum_{k=1}^K \sum_{i=1}^N a_i \mathbf{1}_{y_i=k}} = \frac{a_i}{K} \quad (6)$$

which defines the induced joint training distribution over (x, y) . The corresponding marginal class distribution is:

$$p_1(y=k) = \sum_{i=1}^N p_1(x_i, y_i) \mathbf{1}_{y_i=k} = \frac{\sum_{i=1}^N a_i \mathbf{1}_{y_i=k}}{K} = \frac{1}{K} \quad (7)$$

which means that the induced effective training distribution $p_1(x, y)$ has a uniform class prior (*i.e.*, explicitly class-balanced), regardless of the original class imbalance in $p_0(x, y)$. At the same time, the conditional distribution within each class is: $p_1(x_i|y=k) = a_i$, which is shaped by the learned weights under the simplex constraint. This corresponds to a redistribution of probability mass within each class. Therefore, the mechanism can be understood at two levels: (i) The combination of *WeightedRandomSampler*

²Though the Gaussian assumption may be overly simplistic, CCDB still shows improvements in such cases; extending it to multi-modal modeling is left for future work.

and class-wise simplex constraint enforces a uniform class distribution $p_1(y)$, thereby achieving inter-class balancing and rendering the entropy term redundant. (ii) The simplex constraint determines the intra-class distribution $p_1(x|y)$, emphasizing informative samples within each class.

Thus, the proposed method does not merely perform within-class redistribution; instead, it simultaneously enforces a uniform class distribution and refines the conditional distribution within each class. The final objective for our CCDB is:

$$\begin{aligned} \min_{w_1, \dots, w_K} \quad & \frac{1}{K} \sum_{k=1}^K W_2(z|_{\{y=k, w_k\}}, z) \\ \text{s.t.} \quad & \forall k \in \{1, \dots, K\}, \sum_{i=1}^{n_k} [w_k]_i = 1, \\ & \forall i \in \{1, \dots, n_k\}, [w_k]_i \geq 0, \end{aligned} \quad (8)$$

The optimal solution of Equation (8) reweights samples within each class, emphasizing those scarcely distributed in areas where the other classes are densely clustered, while downweighting the opposite. This aligns the class-conditional distribution with the marginal distribution, ideally achieving $p(z|y=k, w_k) = p(z)$ and independence between z and y . Training on this reweighted dataset effectively removes reliance on spurious factors.

A toy example is provided in Figure 1(right) to demonstrate the effectiveness of our CCDB in addressing three extreme situations where the spurious factors and class labels are either strongly dependent (Figure 1(a)(c)) or completely independent (Figure 1(b)). In Figure 1(a) and (c), 90% of the samples have z_s correlated with class labels, while the remaining 10% display conflicting correlations. For each configuration, we draw the following conclusions: **(a)** Since the data distribution strongly correlates z_s with y , ERM places the decision boundary near the minority groups, leading to poor generalization in that region. In contrast, our CCDB (and stable learning) emphasizes the minority area, resulting in a new distribution with the dependency between z_s and y significantly reduced. Thus the decision boundary is determined solely by z_c . **(b)** When z_s is independent of y , both ERM and CCDB correctly placed the decision boundary. Additionally, CCDB emphasizes samples near the boundary, improving performance in ambiguous regions. **(c)** When predefined features are independent but each is a significant entanglement of core and spurious features (e.g., coincidentally aligned with a Principal Component Analysis (PCA) axis, with each direction being a mixture of z_c and z_s), independent-based stable learning methods fail and place the decision boundary near the minority regions. In contrast, CCDB can still identify and emphasize minority groups, properly correcting the decision boundary.

Algorithm 1 CCDB for group robust classification

Input: Network $f_{\theta_2} \circ f_{\theta_1}$ for biased feature extraction, f_ϕ for classification, logit vectors $\{s_c\}_{k=1}^K$ for sample reweighting, training set $\mathcal{D}_{tr}$ and validation set $\mathcal{D}_v$.

Output: unbiased classifier f_ϕ :

#Stage1: learning biased feature extractor

1: Randomly split a subset from $\mathcal{D}_{tr}$ with proportion γ .

2: Train $f_{\theta_2} \circ f_{\theta_1}$ on the split subset using constrained ERM (Equation (9)).

Stage2: Learning sample weights

3: Extract features $\{z_i\}_{i=1}^{n_k}$ from $\mathcal{D}_{tr}$ using the biased feature extractor f_{θ_1} .

4: Obtain the optimal sample weights $w^* = \{w_1^*, \dots, w_K^*\}$ by solving Equation (8).

Stage 3: Learning unbiased classifier

5: Train classifier f_ϕ on the reweighted samples using standard ERM.

6: Select the best-performing f_ϕ using $\mathcal{D}_v$.

3.2. Full pipeline of our method

With the proposed reweighting mechanism, the complete pipeline of our CCDB is summarized in Algorithm 1, which consists of three stages detailed as follows.

Stage 1: Learning biased feature extractor. In this stage, we train a classifier $f_{\theta_2} \circ f_{\theta_1}$ on $\mathcal{D}_{tr}$, where f_{θ_1} is the backbone and f_{θ_2} is a fully connected classification head. After training, f_{θ_1} is used as a biased feature extractor in the next stage. Intuitively, we aim for the extracted features to be dominated by spurious factors as much as possible, so that their label predictability can be effectively reduced in the subsequent sample reweighting stage. To this end, we introduce the following two strategies:

Data splitting. We experimentally find that the size of the training data significantly affects the dominance of spurious factors. As pointed out in (Pezeshki et al., 2024), when training on the entire training dataset, the model tends to overfit, causing the latent distribution to be less influenced by biased information. This, in turn, makes our method less effective. Figure 3 shows the latent space learned from different splits of the training data. As the proportion of training data, γ , decreases, the extracted features become increasingly dominated by spurious factors. Based on this observation, we propose splitting the training data into a subset with proportion γ (e.g., $\gamma = 50\%$) and training f_{θ_1} on it. The resulting f_{θ_1} is then applied to the entire training set for feature extraction and sample reweighting. Our experiments suggest that data splitting can help expose spurious factors, facilitating more effective balancing during the reweighting process.

Intra-class compactness. In addition to classification, we consider enhancing intra-class compactness in the latent

space by minimizing the distance between samples belonging to the same class. The objective for feature extraction is thus reformed as follows:

$$\begin{aligned} \mathcal{L}_{\theta_1, \theta_2} = & \frac{1}{n} \sum_{i=1}^n l_{\text{CE}}(f_{\theta_2} \circ f_{\theta_1}(\mathbf{x}_i), y_i) \\ & + \frac{\lambda}{K} \sum_{k=1}^K \frac{1}{n_k(n_k-1)} \sum_{i \neq j, y_i=y_j=k} \|f_{\theta_1}(\mathbf{x}_i) - f_{\theta_1}(\mathbf{x}_j)\|^2 \end{aligned} \quad (9)$$

where l_{CE} denotes the cross entropy loss, $\|\cdot\|$ represents the Euclidean norm, λ is used to control the strength of the constraint. In the presence of spurious correlations, certain spurious factors undoubtedly dominate within a specific class, acting as alternative features to $\mathbf{z}_c$ for classification. Imposing intra-class compactness by bringing samples of the same class together serves two purposes (see appendix Figure 8): (i) it clusters data with similar spurious factors, amplifying the bias attributes (samples with the same background tend to cluster together when intra-class compactness is applied); (ii) This smoothing effect encourages the feature space to resemble a Gaussian distribution, which empirically favors the 2-Wasserstein distance used in our method.

Stage 2: Learning sample weights. With the extracted biased features, we compute weights for each training sample by optimizing Equation (8), which effectively reduces the dependence between spurious factors and class labels. For practical implementation, we reparameterize the weight vector $\mathbf{w}_k$ for class k as the softmax of a logits vector $\mathbf{s}_k \in \mathbb{R}^{n_k}$, i.e., $\mathbf{w}_k = \text{softmax}(\mathbf{s}_k)$. This ensures that the weights are non-negative and sum to one, $\sum_{i=1}^{n_k} [\mathbf{w}_k]_i = 1$, with the elements of $\mathbf{s}_k$ being real-valued. To stabilize optimization, we clip the elements of $\mathbf{s}_k$ within the range $[-T, T]$. Please see the experiments and appendix for hyperparameter settings.

Stage 3: Learning unbiased classifier. Two approaches are available for incorporating the learned sample weights into standard ERM for unbiased learning. (i) Weighting samples during data sampling via “`torch.utils.data.WeightedRandomSampler`”. (ii) incorporating the sample weights directly into the cross-entropy loss. We adopt the first approach, as it empirically yields better performance.

4. Experimental results

Dataset. We conduct experiments on five widely used benchmark datasets: cMNIST (Li et al., 2022a), cCIFAR10 (Hendrycks & Dietterich, 2018), Waterbirds (Zhou et al., 2022), CelebA (Sagawa et al.), and CivilComments (Koh et al., 2021) to demonstrate the effectiveness of our method. Among them, cMNIST and cCIFAR10 are ten-way classification tasks, where each class is dominated by a specific color or noise pattern, making them ideal for demonstrating the superior performance of CCDB in challenging multi-

class settings with extremely biased training data. Waterbirds, CelebA, and CivilComments are real-world datasets: Waterbirds and CelebA are image classification datasets where each class is strongly correlated with a specific background or gender attribute; CivilComments is a text classification dataset with the class information spuriously correlated with mentions of certain demographic identities. These datasets are used to showcase the generalizability of CCDB to real-world scenarios. Detailed data information is provided in the supplementary material. For all datasets, we use the standard train-validation-test splits as defined in their respective literature and report both group-balanced test accuracy and the worst-group test accuracy.

Compared baselines. To demonstrate the superiority of our CCDB in addressing distribution shift in the presence of spurious correlations, we compare it with a diverse set of group robustness techniques. GroupDRO is bias-supervised during both training and validation, serving as a challenging baseline. LFF, JTT, LC, and SELF are pseudo-bias-supervised during training but require bias annotations during validation to achieve optimal performance. Similar to our method, standard empirical risk minimization (ERM), uLA, ZIN, MASKTUNE, XRM, and GERNE are bias-agnostic during both training and validation. Reweight, BSoftmax, and ReWeightCRT are baselines designed for class-imbalance.

Training setup. To make a fair comparison, we apply a 3-hidden layer MLP for cMNIST, a ResNet18 for cCIFAR10, a ResNet50 for Waterbirds and CelebA, and a BERT for CivilComments following existing methods (Tsirigotis et al., 2023). Both ResNet18 and ResNet 50 are pretrained on ImageNet-1K, BERT is pretrained on Book Corpus and English Wikipedia. The proposed CCDB method consists of three stages, each optimized independently. The first two training stages span 20 epochs and 1000 iterations across all datasets (except CivilComments with 10 epochs); the third stage involves 2000 iterations for Waterbirds and CelebA, 5000 iterations for cMNIST and cCIFAR10, and 10 epochs for CivilComments.

On data preprocessing, we apply simple augmentations to each dataset following (Ahuja et al., 2021), designed not to induce invariance to variations in the bias attribute. Consequently, the improvements reported can be attributed to the proposed CCDB methodology rather than bias-nullifying data augmentations. All the results are averaged over 5 random seeds. See supplementary material for the detailed experimental setup.

4.1. Experiments on multi-class data

In this section, we use multi-class datasets (i.e., cMNIST and cCIFAR10) with controllable synthetic bias to evaluate the effectiveness of our method in mitigating varying levels of bias under challenging multi-class conditions. For

Table 1. Results on cMNIST and cCIFAR10 with various bias-conflicting ratios in the training set. The test accuracy(%) is averaged over 5 random seeds. The best results are indicated in bold.

Methods	Bias label		cMNIST				cCIFAR10			
	Train	Val	0.5%	1%	2%	5%	0.5%	1%	2%	5%
GroupDRO(Sagawa et al., 2020)	Yes	Yes	63.12	68.78	76.30	84.20	33.44	38.30	45.81	57.32
LFF (Nam et al., 2020)	No	Yes	52.50 $\pm$ 2.43	61.89 $\pm$ 4.97	71.03 $\pm$ 2.44	80.57 $\pm$ 3.84	28.57 $\pm$ 1.30	33.07 $\pm$ 0.77	39.91 $\pm$ 0.30	50.27 $\pm$ 1.56
LC (Liu et al., 2023)	No	Yes	71.25 $\pm$ 3.17	82.25 $\pm$ 2.11	86.21 $\pm$ 1.02	91.16 $\pm$ 0.97	34.56 $\pm$ 0.69	37.34 $\pm$ 0.69	47.81 $\pm$ 2.00	54.55 $\pm$ 1.26
ERM	No	No	35.19 $\pm$ 3.49	52.09 $\pm$ 2.88	65.86 $\pm$ 3.59	82.17 $\pm$ 0.74	23.08 $\pm$ 1.25	25.82 $\pm$ 0.33	30.06 $\pm$ 0.71	39.42 $\pm$ 0.64
uLA (Tsirigotis et al., 2023)	No	No	75.13 $\pm$ 0.78	81.80 $\pm$ 1.41	84.79 $\pm$ 1.10	92.79 $\pm$ 0.85	34.39 $\pm$ 1.14	62.49 $\pm$ 0.74	63.88 $\pm$ 1.07	74.49 $\pm$ 0.58
GERNE (Asaad et al., 2025)	No	No	77.25 $\pm$ 0.17	83.98 $\pm$ 0.26	87.41 $\pm$ 0.31	90.98 $\pm$ 0.13	39.90 $\pm$ 0.48	45.60 $\pm$ 0.23	50.19 $\pm$ 0.18	56.53 $\pm$ 0.32
CCDB	No	No	83.20$\pm$2.17	87.95$\pm$1.59	91.02$\pm$0.28	96.37$\pm$0.25	55.07$\pm$0.85	63.28$\pm$0.46	67.78$\pm$0.78	74.64$\pm$0.34

Table 2. Results on Waterbirds, CelebA and CivilComments. We report average test accuracy(%) and std.dev. over 5 random seeds.

Methods	Bias label		Waterbirds		CelebA		CivilComments	
	Train	Val	i.i.d.	WorstGroup	i.i.d.	WorstGroup	i.i.d.	WorstGroup
GroupDRO (Sagawa et al., 2020)	Yes	Yes	93.50	91.40	92.90	88.90	84.2	73.7
LFF (Nam et al., 2020)	No	Yes	97.50	75.20	86.00	77.20	68.2	50.3
JTT (Liu et al., 2021)	No	Yes	93.60	86.00	88.00	81.10	83.3	64.3
LC (Liu et al., 2023)	No	Yes	-	90.50 $\pm$ 1.1	-	88.10 $\pm$ 0.8	-	70.30 $\pm$ 1.2
SELF (LaBonte et al., 2023)	No	Yes	-	93.00 $\pm$ 0.3	-	83.90 $\pm$ 0.9	-	79.10 $\pm$ 2.1
ReWeight (Japkowicz, 2000)	No	Yes	86.9 $\pm$ 0.50	72.50 $\pm$ 0.30	92.00 $\pm$ 0.40	81.50 $\pm$ 0.90	83.10 $\pm$ 0.10	69.60 $\pm$ 0.60
BSoftmax (Ren et al., 2020)	No	Yes	88.4 $\pm$ 1.20	74.10 $\pm$ 0.90	91.90 $\pm$ 0.12	83.30 $\pm$ 0.5	83.00 $\pm$ 0.40	69.40 $\pm$ 1.20
ReWeightCRT (Kang et al.)	No	Yes	89.40 $\pm$ 0.30	76.30 $\pm$ 0.20	94.20 $\pm$ 0.10	71.10 $\pm$ 0.50	83.40 $\pm$ 0.00	68.20 $\pm$ 0.40
ERM	No	No	97.30	72.60	95.60	47.20	81.6	66.7
ZIN (Lin et al., 2022)	No	No	-	74.58	-	67.27	-	-
MASKTUNE (Asgari Taghanaki et al., 2022)	No	No	93.00 $\pm$ 0.7	86.40 $\pm$ 1.9	91.30 $\pm$ 0.1	78.00 $\pm$ 1.2	-	-
uLA (Tsirigotis et al., 2023)	No	No	91.50 $\pm$ 0.7	86.10 $\pm$ 1.5	93.90 $\pm$ 0.2	86.50 $\pm$ 3.7	-	-
XRM (Pezeshki et al., 2024)	No	No	90.60	86.10	91.0	88.5	83.5	70.1
GERNE (Asaad et al., 2025)	No	No	-	89.88 $\pm$ 0.67	-	74.24 $\pm$ 2.51	-	63.10 $\pm$ 0.22
CCDB	No	No	92.59 $\pm$ 0.10	90.48$\pm$0.28	90.08 $\pm$ 0.19	85.27 $\pm$ 0.28	83.60$\pm$0.21	75.00$\pm$0.26

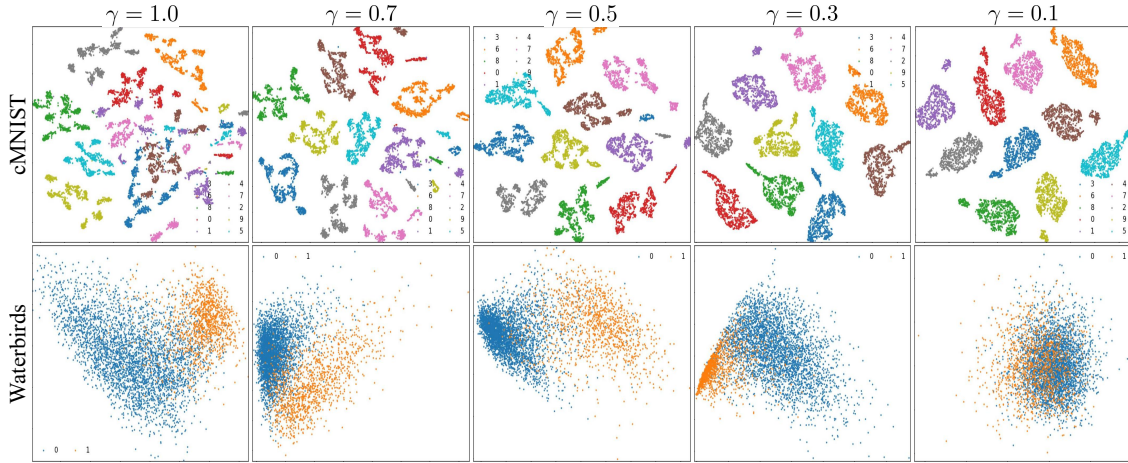


Figure 2. t-SNE visualization of the latent space learned in Stage1. Colored by bias annotations. As γ decreases, the feature space exhibits an increasingly pronounced clustering effect aligned with bias annotations.

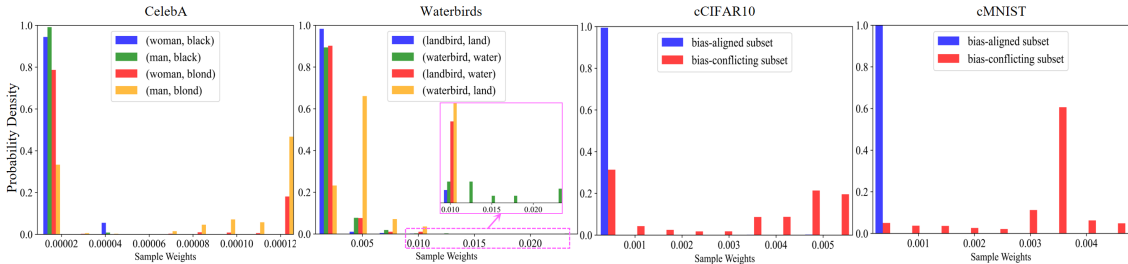


Figure 3. Distribution of the learned sample weights within each group. The distributions are normalized within each group.

each dataset, we control the degree of spurious correlation by varying the proportion of bias-conflicting samples in the training data, and evaluate performance on completely

unbiased test set. Following (Tsirigotis et al., 2023), the bias-conflicting ratios are set to $\{0.5\%, 1\%, 2\%, 5\%\}$ for both datasets, with 0.5% representing an extremely biased

scenario. The results are reported in Table 1. From which, we have the following observations: (i) On cMNIST, our method consistently achieves the best performance, surpassing the second best with a large margin, even though some competing methods utilize bias annotations during model training or selection; (ii) On cCIFAR10, our method performs the best on average and significantly outperforms the others when the bias-conflicting ratio is as low as 0.5%. Considering that uLA employs resource-consuming MO-COV2+ (Chen et al., 2020) pretraining and broad hyperparameter searching, our method provides a favorable trade-off between performance and computational complexity.

4.2. Experiments on real-world data

Experiments on real-world datasets, Waterbirds, CelebA, and CivilComments are conducted to test the effectiveness of our method in dealing with spurious correlations in practical scenarios. The results are reported in Table 2. With the supervision of bias annotations, GroupDRO demonstrates strong generalization performance on the worst group. On all three datasets, our method achieves competitive or superior performance compared to those utilizing bias information during validation. Compared with bias-agnostic methods, our method achieves performance comparable to uLA on CelebA and to GERNE on Waterbirds, while significantly outperforming existing approaches in the remaining settings. Notably, although the stable learning method ZIN improves over ERM, its performance remains below recent debiasing methods, possibly due to its reliance on global invariance constraints without explicitly modeling spurious features. By referring to Table 1, we observe that bias supervision is more effective for simple binary classification tasks, *e.g.*, GroupDRO achieves SOTA performance on Waterbirds and CelebA. However, for more complex multi-class classification tasks, bias-supervised methods fall significantly behind bias-free approaches (*e.g.*, uLA and our CCDB).

4.3. Ablation study

We conduct an ablation study to guide hyperparameter selection without bias annotations. Although empirical, the settings achieve strong performance, and we will explore more robust strategies in future work.

The proportion γ split from the training data for biased feature extractor learning. By varying γ in $\{10\%, 30\%, 50\%, 70\%, 100\%\}$, we visualize the t-SNE of the learned feature space during stage1 in Figure 2 and report the worst group accuracy of stage 3 in Figure 4 (left). In Figure 2, for cMNIST, as γ decreases, the feature space exhibits an increasingly pronounced clustering effect aligned with bias annotations, indicating a feature space dominated by spurious factors. For Waterbirds, the dominance of spurious factors initially increases but then diminishes when

$\gamma = 0.1$, likely due to the smaller scale of the training data. Accordingly, with an appropriately small $\gamma \in [0.3, 0.5]$, spurious factors are more effectively identified and balanced, leading to a classifier that relies more on core factors and exhibiting improved generalization capabilities, see the optimal test accuracy in Figure 4 (left) for verification.

The threshold T for sample weight clipping. We evaluate the impact of T by varying it from 0.5 to 3.0, with the results shown in Figure 4(left). For comparison, we also report the performance without clipping (“No clipping”), demonstrating that unbounded weights introduce excessive flexibility, which leads to over-weighting minority groups and significantly degrades overall performance. The threshold T is closely related to the number of classes. For datasets with fewer classes, aligning each class-conditional distribution with the marginal distribution is easier and therefore requires a smaller T (*e.g.*, $T \approx 1.5$ on Waterbirds). In contrast, datasets with more classes require a relatively larger T to effectively reduce the distribution discrepancy (*e.g.*, $T \approx 2.5$ on cCIFAR10).

The effectiveness of the mutual information (MI) term. To show that MI in CCDB provides gains beyond class balancing, we compare it with baselines designed for class-imbalance (Reweight, BSoftmax, ReWeightCRT) in Table 2. CCDB consistently surpass these methods with large margin, indicating that its advantage stems from jointly addressing both challenges.

4.4. Analysis on sample weights learned by CCDB

To show that the proposed sample reweighting technique effectively reduces the correlation between spurious factors z_s and class labels y , we analyze the distribution of the learned sample weights and their impact on the spuriousness of the features extracted for classification. Figure 3 shows the histograms of sample weights obtained by CCDB. For CelebA and Waterbirds, the minority groups (man-blond, waterbird-land, landbird-water) receive weights roughly ten times higher than the majority groups, reflecting our primal goal of up-weighting underrepresented samples. For cCIFAR10 and cMNIST, we divide the data into a bias-aligned subset, where z_s aligns with y , and a bias-conflicting subset, where z_s conflicts with y . The weights for the minority conflicting subset are concentrated at relatively high values, while the aligned subset has weights close to zero.

Figure 4(right) shows the effect of sample reweighting on the spuriousness of the learned features. We measure spuriousness by calculating the Pearson correlation coefficient (Benesty et al., 2009) between each feature dimension and class/bias information, and present a boxplot of these coefficients. Before reweighting, most extracted feature dimensions are dominated by spurious factors, showing high correlation with bias and low correlation with labels (blue

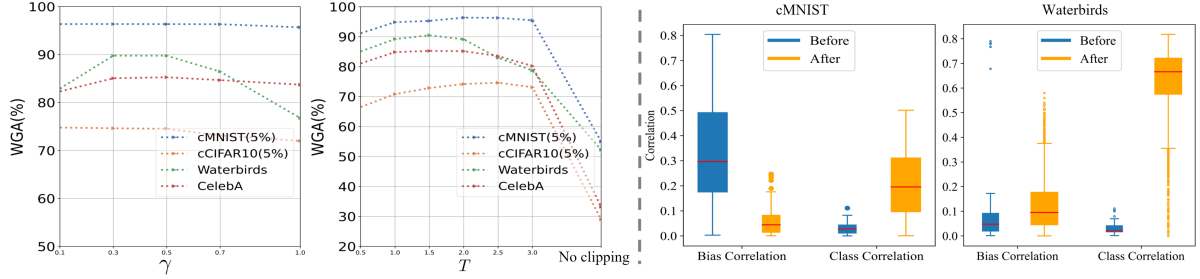


Figure 4. **Left:** Ablation study on the split ratio γ and threshold T . **Right:** Boxplot of the feature correlations with bias and class information, before and after CCDB sample reweighting. Refer to appendix for more results.

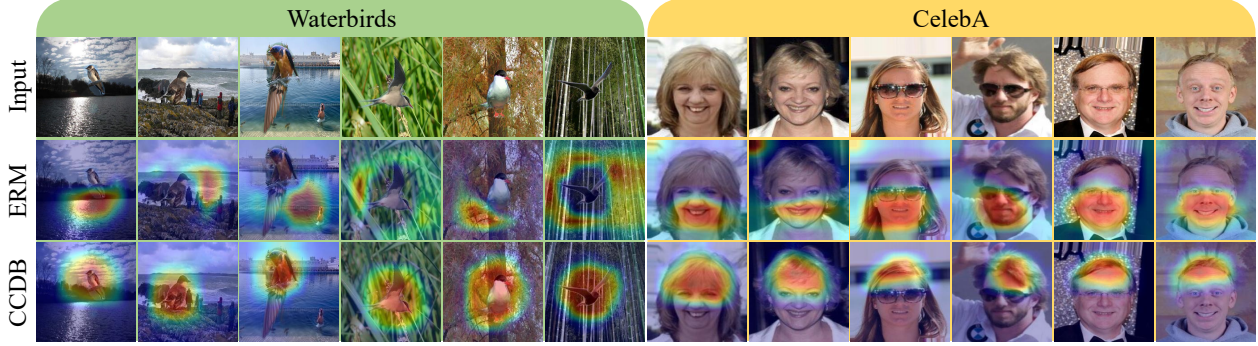


Figure 5. Visualization of ERM and CCDB attention on Waterbirds and CelebA samples.

boxes). After CCDB reweighting, the features become predominantly core, strongly correlated with class labels and weakly correlated with bias (yellow box). These results highlight the effectiveness of CCDB in mitigating spurious correlations, yielding a de-biased data distribution for downstream tasks.

To further demonstrate the effectiveness of the proposed CCDB in getting rid of spurious correlations, we compare the classifier’s attention before and after sample reweighting in Figure 5 (visualized via xGradCAM (Selvaraju et al., 2017)). Without sample reweighting, the model obtained from ERM primarily focuses on spurious factors, *e.g.*, background or facial details. In contrast, after applying CCDB reweighting, the model is enforced to explore core factors, *e.g.*, the foreground birds and hair color, leading to more robust predictions.

5. Conclusion

In this work, we considered the challenge problem of preventing models from learning spurious correlations without relying on bias annotations. We provide a new perspective on the cause of spurious correlations: the imbalance in both class-conditional and class-marginal distributions. Accordingly, we propose a novel class-conditional distribution balancing (CCDB) technique that effectively removes spurious correlations from the training data via a modified class-specific sample reweighting mechanism. The proposed CCDB performs exceptionally well in both basic

binary and challenging multi-class scenarios without relying on bias/group predictions or expensive pretrained foundation models, offering a novel and promising direction for mitigating spurious correlations. While our method demonstrates strong empirical performance, several directions remain for future work, including improving robustness under highly entangled feature distributions, extending the current Gaussian approximation to more complex multimodal settings, and developing more adaptive weighting strategies for noisy or challenging samples (see appendix for further discussion).

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Impact Statement

This paper presents work whose goal is to advance the field of machine learning. While our method improves worst-group robustness on standard benchmarks, it is not uniformly optimal across all settings, and improvements in worst-group accuracy do not necessarily translate directly into fairness in real-world deployment, especially when minority patterns are noisy or underrepresented. This reflects the inherent trade-off between robustness and average ac-

curacy, as well as the gap between benchmark robustness and real-world fairness. Therefore, caution should be exercised when applying the proposed method in high-stakes domains, where fairness and robustness requirements may differ substantially from the assumptions underlying standard benchmarks.

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Table 3. An illustration of the datasets used in this paper. In the training data, z_s is highly correlated with the class labels. However, this correlation is either broken or even reversed in the test set.

Dataset	z_c	z_s	classes	Train	Val	Test
cMNIST	digit	digit color	10	95% $\sim$ 99.5% biased	95% $\sim$ 99.5% biased	unbiased
cCIFAR10	object	noise	10	95% $\sim$ 99.5% biased	95% $\sim$ 99.5% biased	unbiased
Waterbirds	bird	background	2	{3498, 184, 56, 1057}	{467, 467, 133, 133}	{2225, 2225, 642, 642}
CelebA	hair color	gender	2	{71629, 66874, 22880, 1387}	{8535, 8276, 2874, 182}	{9767, 7535, 2480, 180}
CivilComments	toxic	identity	2	{148186, 90337, 12731, 17784}	{25159, 14966, 2111, 2944}	{74780, 43778, 6455, 8769}

Table 4. The hyperparameters and augmentations used during the training of our CCDB.

Dataset	Learning rate	Batch size	Weight decay	{Epoch, Iter, Iter}	λ	Split γ	Clip T	Augmentation
cMNIST	1e-2	256	1e-4	{5, 1000, 5000}	0.5	0.3	2	None
cCIFAR10	1e-4	256	1e-4	{5, 1000, 5000}	0.5	0.3	2	Resized crop, horizontal flip
Waterbirds	1e-5	256	1e-4	{20, 1000, 2000}	0.1	0.5	1.5	Resized crop, horizontal flip
CelebA	1e-5	256	1e-4	{20, 1000, 2000}	0.1	0.5	1.5	Resized crop, horizontal flip
CivilComments	1e-5	16	1e-4	{10, 1000, 10epoch}	0.1	0.5	1.5	None

A. Related works

We categorize prior literature according to whether bias information is used to mitigate spurious correlations, as detailed below.

A.1. Bias-guided methods

Addressing spurious correlations becomes straightforward when bias annotations are available. By combining both bias and label annotations, one can create a group balanced training process or model selection strategy, to enhance robustness against spurious correlations. For example, Group distributionally robust optimization (GroupDRO) (Sagawa et al.; Duchi et al., 2021; Oren et al., 2019; Hu et al., 2018) proposes to minimize the worst-group loss based on predefined group assignments. To alleviate the reliance on bias annotations, deep feature reweighting (DFR) (Kirichenko et al., 2022; Izmailov et al., 2022) proposes retraining only the last layer of the model using a small, group-balanced held-out dataset. MAPLE (Zhou et al., 2022) utilizes a validation set with bias annotations to reweight the training samples, ensuring that the training process focuses on the worst group. Pseudo-bias-supervised methods propose to use the predicted bias information instead of explicit annotations, typically through auxiliary models. Learning from Failure (LFF) (Nam et al., 2020), logit adjustment technique (LC) (Liu et al., 2023) and Disentangled Feature Augmentation (DFA) (Chu et al., 2021) reweight the loss of the unbiased model using a co-trained biased model. Meanwhile, Just-train-twice (JTT) (Liu et al., 2021) reweights misclassified samples from an initial biased training phase, emphasizing data from the worst-performing groups. Additionally, research in (Kim et al., 2024; Yu et al., 2024; Clark et al., 2019) integrates large-scale vision-language models for more precise bias prediction. However, the performance of these methods depends on the prediction quality and still requires bias annotations during validation for optimal performance.

A.2. Bias-agnostic methods

To mitigate the dependence on bias annotations/predictions, researchers have developed bias-agnostic methods under different motivations. Stable learning based methods (Kuang et al., 2018) propose to decorrelate all features to exclude spurious correlations. For example, (Kuang et al., 2020) proposes a sample reweighting approach to decorrelate input variables and demonstrate stable performance across distribution shifts. StableNet (Zhang et al., 2021) further generalizes these techniques to nonlinear feature decorrelation. However, achieving perfect decorrelation theoretically requires a sufficiently large dataset, which is often impractical. To compensate for the imperfectness of sample reweighting under finite-sample settings, Sparse variable independent(SVI) (Yu et al., 2023) introduces a sparsity constraint to avoid unnecessary decorrelation within core

variables. ZIN (Lin et al., 2022) extends stable learning by learning invariant representations without requiring explicit environment partition, aiming to suppress unstable correlations through invariance learning rather than feature decorrelation. Nevertheless, most of these methods are either developed under linear or shallow frameworks with predefined features, or remain limited in effectively separating core and spurious features in deep models. This is because representations learned by deep models are often heavily entangled, making decorrelation or invariance constraints alone insufficient for reliably separating core and spurious factors (Yu et al., 2023). Another line of research utilizes disagreements among diverse models to improve generalization under distribution shifts. For instance, Diversity by disagreement (Pagliardini et al., 2023) and Diversify and disambiguate (Lee et al., 2022) propose to maximize the disagreement between multiple predictors to learn a diverse ensemble, which require a completely spuriously correlated training dataset. Selective last-layer finetuning (SELF) (LaBonte et al., 2023) proposes reweighting the dataset based on misclassifications or disagreements observed before and after last-layer retraining. However, these methods still need a small set of group annotations for model selection. Several other bias-agnostic methods have demonstrated competitive performance, *e.g.*, General Greedy De-bias Learning (Han et al., 2023) leverages both a biased model and a base model to assess the difficulty of training samples, thereby focusing training on the more challenging ones. However, this approach has difficulty balancing minority- and majority-group performance, necessitating the use of curriculum regularization to strike a better balance. uLA (Tsirigotis et al., 2023) employs biased prediction to adjust the logits for debias learning. It relies heavily on extensively pretrained self-supervised models, which significantly influence the final performance. XRM (Pezeshki et al., 2024) constructs twin networks to fully predict bias across the entire training dataset, using a label-flipping mechanism to intentionally steer the model toward biased predictions. MASKTUNE (Asgari Taghanaki et al., 2022), on the other hand, explores new features during a secondary training phase by masking previously identified spurious factors. However, it requires careful selection of the masking threshold to ensure stable performance. GERNE (Asaad et al., 2025) identifies a debiasing direction by contrasting gradients computed from two distinct batches and updates the model accordingly. The work presented in (Yang et al., 2023) introduces a novel perspective on the causes of spurious correlations. While it shares certain similarities with our approach, it is grounded in a different framework and primarily aims to offer a comprehensive analysis of existing methods.

Our method also falls within the bias-agnostic framework. In contrast to the aforementioned methods, our approach (i) does not decorrelate all features, but instead takes a more straightforward approach by maximizing the conditional entropy of label information given the spurious factors; (ii) allows for easy control of sample weighting through distribution distance minimization.

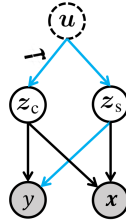


Figure 6. A detailed causal graph.

B. Viewing CCDB from a causal inference perspective

The causal graph in Figure 1 (left) of the main manuscript is constructed following (Neal, 2020; Yin et al., 2024). A more detailed version is shown in Figure 6, where blue arrows represent paths whose parameters vary across domains, while black arrows denote domain-invariant paths. Here, u denotes an unobserved confounder (*e.g.*, the domain or environment) that simultaneously affects both z_c and z_s , and this causal influence changes across domains. The mechanism by which z_c and z_s jointly generate the observation x is assumed to be invariant across domains. The prediction of y based on z_c is also domain-invariant, whereas the prediction of y based on z_s varies across domains. Treating z_c as the treatment and y as the outcome induces a causal path $z_c \rightarrow y$ as well as a backdoor path $z_c \leftarrow u \rightarrow z_s \rightarrow y$. The spurious correlation between z_s and y arises from the unknown confounder u . To safely estimate the causal effect of an image feature z on the label y , it is necessary to remove the confounding bias induced by the backdoor path and directly estimate the causal effect of z_c on y , which remains invariant across domains, namely,

$$p(y|do(z_c)) = \mathbb{E}_{p(z_s)} p(y|z_c, z_s) \quad (10)$$

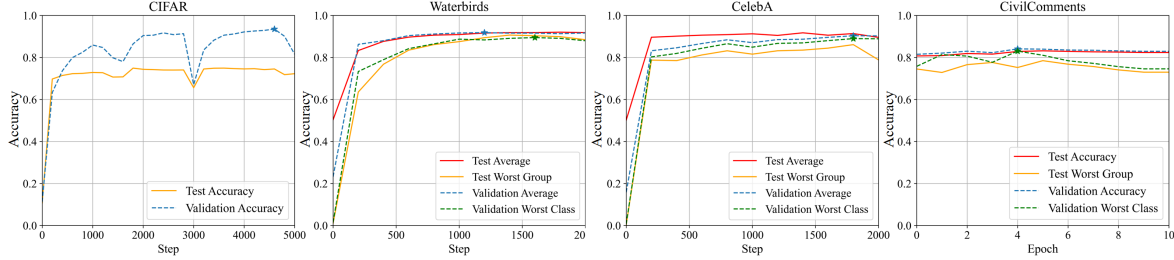


Figure 7. The training process of CCDB on cCIFAR10, Waterbirds, CelebA, and CivilComments. Both the standard average accuracy and the worst-class accuracy on the validation set are effective for selecting models with strong test performance. The best validation accuracy is marked with a star.

where $do(z_c)$ denotes the do-operator from causal inference, which cuts off all incoming paths to z_c , thereby removing the spurious correlation between z_c and z_s , *i.e.*, $p(z_s|z_c) = p(z_s)$, a condition that reflects covariate balance. Given that the core factor z_c and the label y have a strong correlation, it follows that $p(z_s|y) \approx p(z_s)$. This condition is approximate to the optimal solution that minimizes the mutual information term in our CCDB framework.

C. Data

We conduct experiments on five widely used benchmark datasets: cMNIST (Li et al., 2022a), cCIFAR10 (Hendrycks & Dietterich, 2018), Waterbirds (Zhou et al., 2022), CelebA (Sagawa et al.), and CivilComments (Koh et al., 2021) to demonstrate the effectiveness of our method. The detailed statistics of each dataset are summarized in Table 3. Among these datasets, cMNIST and cCIFAR10 are ten-way classification tasks in which each class is dominated by a specific color or noise pattern, with varying bias ratios. Waterbirds, CelebA, and CivilComments are real-world datasets: Waterbirds and CelebA are image classification datasets where each class is strongly correlated with a particular background or gender attribute, respectively; CivilComments is a text classification dataset where the label is spuriously correlated with mentions of certain demographic identities.

D. Training setup

To make a fair comparison, we apply a 3-hidden layer MLP for cMNIST, a ResNet18 for cCIFAR10, a ResNet50 for Waterbirds and CelebA, and a BERT for CivilComments following existing methods (Tsirigotis et al., 2023; Liu et al., 2021; Pezeshki et al., 2024). Both ResNet18 and ResNet 50 are pretrained on ImageNet-1K, BERT is pretrained on Book Corpus and English Wikipedia. The proposed CCDB method consists of three stages, each optimized independently. The first two training stages span 20 epochs and 1000 iterations across all datasets (except CivilComments with 10 epochs); the third stage involves 2000 iterations for Waterbirds and CelebA, 5000 iterations for cMNIST and cCIFAR10, and 10 epochs for CivilComments. For Stage1, we use the Adam optimizer with a learning rate of 2×10^{-4} and default momentum parameters $\beta = (0.9, 0.999)$ for all image datasets and AdamW optimizer with a learning rate of 1×10^{-5} for the NLP dataset; For stage2, the Adam optimizer with a learning rate of 1×10^{-2} is adopted across all datasets; The learning rate, along with weight decay and batch size used in stage3 for each dataset are detailed in Table 4.

For numerical stability, covariance matrices are estimated from mini-batches, and diagonal regularization ($\epsilon \mathbf{I}$, $\epsilon = 10^{-5}$) is applied during matrix square-root computations. Matrix square roots are computed via eigen-decomposition with eigenvalue clipping using the same threshold $\epsilon = 10^{-5}$. On data preprocessing, we apply simple augmentations to each dataset as listed in Table 4, following (Ahuja et al., 2021). These augmentations are specifically designed not to induce invariance to variations in the bias attribute. Consequently, the improvements reported in the main paper can be attributed to the proposed CCDB methodology, rather than to the use of bias-nullifying data augmentations.

All the results are averaged over 5 random seeds, with the best-performing model (highest validation accuracy) selected for each seed. From the results in Figure 7, we conclude that CCDB can use either standard average accuracy or worst-class accuracy on a class-balanced validation set for model selection.

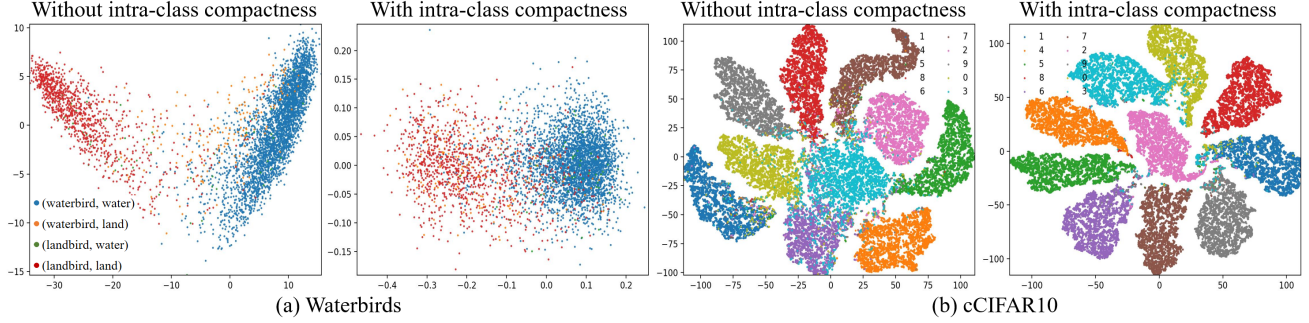


Figure 8. The effect of intra-class compactness on the latent space. t-SNE visualization of the latent space learned on: (a) Waterbirds colored by group annotations; and (b) cCIFAR10 Colored by the spurious noise type attribute. With intra-class compactness, the distribution demonstrates a strong alignment with spurious information.

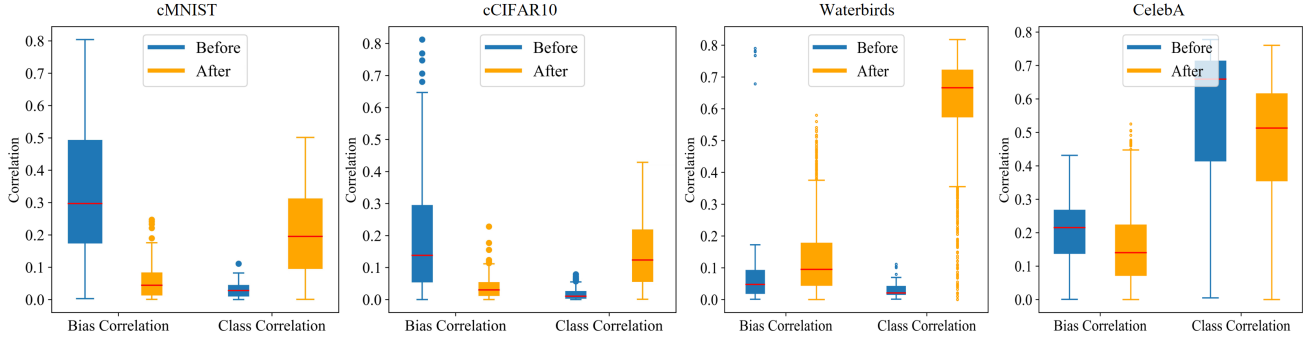


Figure 9. Boxplot of the feature correlations with bias and class information, before and after CCDB sample reweighting.

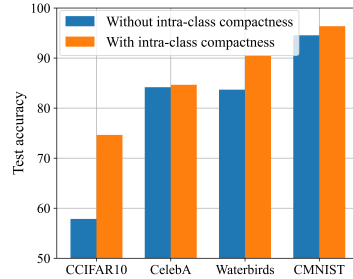


Figure 10. The effect of intra-class compactness on the final performance of CCDB, with the worst group accuracy for Waterbirds and CelebA, and standard average accuracy for cMNIST and cCIFAR10.

E. Additional experimental results

E.1. On model selection for CCDB

To demonstrate that standard average accuracy on the validation set serves as an effective model selection criterion for CCDB, we record its training process on four datasets (cCIFAR10, Waterbirds, CelebA, and CivilComments), as shown in Figure 7. The average and worst-group accuracy curves on the validation set exhibit similar trends throughout the training process, with their optimal values occurring in regions where the test worst-group performance remains relatively high. This alignment represents an additional advantage of our method.

E.2. The computational cost of stage 2

The stage 2 of CCDB processes all training samples in a single batch, which requires additional storage for the latent representations of the training set. However, the computational cost of this stage is low and acceptable for the following reasons: (i) the process operates directly in the latent space, requiring only a tensor of size $N \times D$ to be stored, where N is

the number of training samples and D is the dimension of the latent representation. (ii) The optimization involves only the calculation of means and covariances, both of which can be performed efficiently.

For optimization of Equation (8), we use Adam with learning rate 0.01. Table 5 shows the storage and running time on all datasets. Obviously, the optimization is overall fast within a few minutes.

Table 5. Computation resource required in Stage2 on four datasets with a single GPU A40.

Cost	cMNIST	cCIFAR10	Waterbirds	CelebA
Storage(M)	6.71	22.00	2.34	79.48
Time(s)	80.45	114.03	39.21	232.14

E.3. The effect of intra-class compactness

The intra-class compactness constraint is applied to amplify bias attributes and smooth the feature space, thereby enhancing the model’s generalization ability. In this section, we conduct an ablation study on cMNIST, cCIFAR10, Waterbirds, and CelebA to evaluate the impact of intra-class compactness on the final performance of CCDB.

The results are presented in Figures 8 and 10. In Figure 8, under naive ERM training (without the intra-class compactness constraint), the feature distribution is skewed and primarily aligned with class information. In contrast, when intra-class compactness is applied, the distribution shifts to align more with bias attributes (such as background or noise type). Additionally, the structure of each cluster more closely resembles a Gaussian distribution compared to the results without intra-class compactness. Figure 10 shows the average and worst group accuracy with and without the intra-class compactness constraint. We observe that, for large-scale (*e.g.*, CelebA) and simpler datasets (*e.g.*, cMNIST), the constraint has a relatively modest effect on the final performance. However, for small-scale (*e.g.*, Waterbirds) and complex (*e.g.*, cCIFAR10) datasets, the constraint significantly improves performance, with gains of up to 16% in accuracy.

We use parameter λ to control the strength of intra-class compactness. Empirically, we find that λ has only a minor impact on the final performance (see Figure 11 (a)). Upon closer inspection, larger values of λ tend to be preferred for simpler, low-dimensional datasets (*e.g.*, cCIFAR10), while smaller values work better for more complex datasets (*e.g.*, Waterbirds). Overall, choosing $\lambda \in [0.1, 0.5]$ yields robust performance.

E.4. Analysis on sample weights learned by CCDB

Figure 9 shows the effect of sample reweighting on the spuriousness of the learned features. We measure spuriousness by calculating the Pearson correlation coefficient (Benesty et al., 2009) between each feature dimension and class/bias information, and present a boxplot of these coefficients. Note, the coefficients are computed on the test data, where no spurious correlation exists, meaning that class labels are solely correlated with core factors and independent of spurious ones. Before applying sample reweighting, most extracted feature dimensions from ERM are dominated by spurious factors, exhibiting high correlation with bias information and low correlation with class labels (blue box). In contrast, after CCDB reweighting, the extracted features are predominantly core, with strong correlation to class labels and weak correlation to bias information (yellow box). These observations highlight the effectiveness of CCDB in mitigating spurious correlations, thus creating a de-biased data distribution for downstream classification tasks.

E.5. Ablation study on the role of mutual information

To disentangle the contribution of the mutual information (MI) from the role of the initially biased model, we conduct an additional ablation study as follows.

Specifically, we compare CCDB with two reweighting baselines use the same biased model to identify high-loss samples, but replace the MI with simple up-weighting: (i) Upweight-group, which partitions samples into majority/minority groups using a loss threshold; (ii) Upweight-loss, which directly assigns sample weights proportional to their loss. For both baselines, we perform hyperparameter tuning and report the best results in Table 6 top.

While both reweighting strategies improve over vanilla ERM, they consistently underperform CCDB. This demonstrates that the MI-based class-conditional distribution balancing provides benefits beyond merely identifying hard or minority samples via a biased model.

Table 6. Compared with class balanced methods.

Methods	Bias label		Waterbirds		CelebA		CivilComments	
	Train	Val	i.i.d.	WorstGroup	i.i.d.	WorstGroup	i.i.d.	WorstGroup
ERM [†]	No	No	97.30	72.60	95.60	47.20	81.6	66.7
Upweight-group	No	No	90.76 $\pm$ 0.24	85.01 $\pm$ 0.41	90.06 $\pm$ 0.12	80.56 $\pm$ 0.16	82.01 $\pm$ 0.30	70.45 $\pm$ 0.17
Upweight-loss	No	No	90.03 $\pm$ 0.41	84.78 $\pm$ 0.54	89.01 $\pm$ 0.25	81.14 $\pm$ 0.34	82.84 $\pm$ 0.35	71.05 $\pm$ 0.34
CCDB	No	No	92.59 $\pm$ 0.10	90.48 $\pm$ 0.28	90.08 $\pm$ 0.19	85.27 $\pm$ 0.28	83.60 $\pm$ 0.21	75.00 $\pm$ 0.26
ReWeight	No	Yes	86.9 $\pm$ 0.50	72.50 $\pm$ 0.30	92.00 $\pm$ 0.40	81.50 $\pm$ 0.90	83.10 $\pm$ 0.10	69.60 $\pm$ 0.60
BSoftmax	No	Yes	88.4 $\pm$ 1.20	74.10 $\pm$ 0.90	91.90 $\pm$ 0.12	83.30 $\pm$ 0.5	83.00 $\pm$ 0.40	69.40 $\pm$ 1.20
ReWeightCRT	No	Yes	89.40 $\pm$ 0.30	76.30 $\pm$ 0.20	94.20 $\pm$ 0.10	71.10 $\pm$ 0.50	83.40 $\pm$ 0.00	68.20 $\pm$ 0.40

To show that the mutual information term in CCDB contributes to performance beyond class balancing, we compare it with baselines designed for class-imbalance (Reweight, BSoftmax, ReWeightCRT (Yang et al., 2023)) in Table 6. CCDB consistently outperforms these methods, suggesting its advantage comes from jointly addressing both challenges. In contrast, methods focusing only on class balancing perform substantially worse (Table 6, bottom).

F. Additional questions

F.1. Concern about Gaussian assumption and 2-Wasserstein distance.

While we agree that deep feature distributions can be complex, multi-modal, and non-Gaussian, we believe that the Gaussian assumption serves as a practical and effective approximation rather than a limiting factor. Our method does not assume strict Gaussianity; instead, we adopt a Gaussian approximation to enable tractable optimization. This is a standard practice in deep learning (*e.g.*, VAE, FID), where matching first- and second-order statistics has proven effective in practice.

Importantly, our objective is not to fully match complex geometries, but to align dominant statistics across classes. Empirically, we observe consistent improvements across diverse datasets, suggesting that this approximation is sufficient to mitigate major spurious correlations.

We agree that extending the method to mixtures of Gaussians is a promising direction, and we leave a more comprehensive study for future work. As an initial exploration, we model multi-modality within each class using a hard confusion matrix and perform distribution matching at the sub-mode level. Results in Table 7(2nd row) show almost consistent improvements in worst-group accuracy (WGA).

We considered an alternative choices for 2-Wasserstein distance, symmetric KL, which relies on moment matching under Gaussian assumptions but introduce additional challenges such as asymmetry and instability. Results in Table 7(last row) show almost comparable performance, further indicating that the choice of distance is not the primary factor.

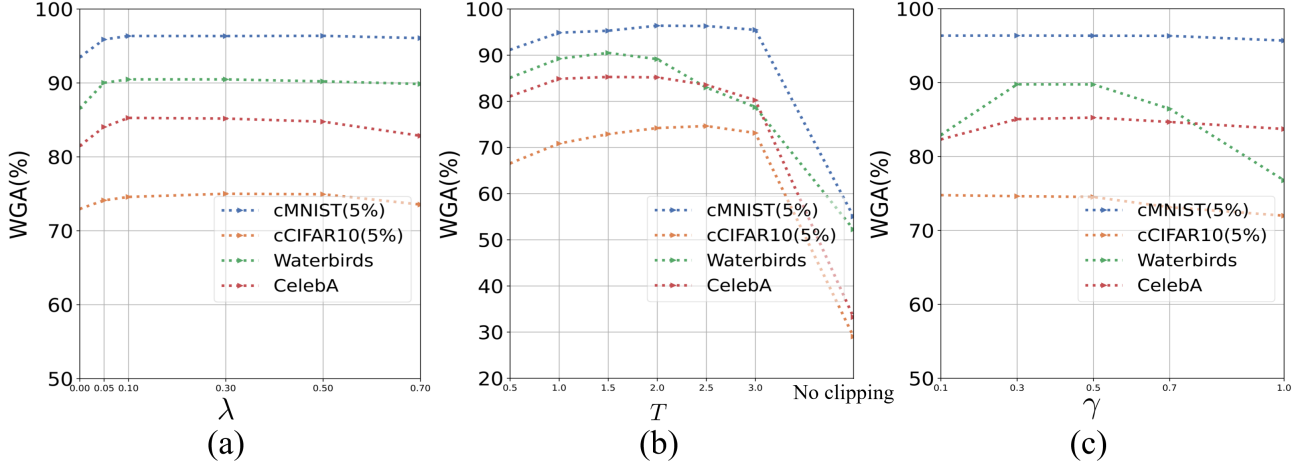
Table 7. Our method with different Gaussian assumptions or different distance metrics.

	Waterbirds	cMNIST(5%)	cCIFAR10(5%)
single Gaussian	90.48	96.37	74.64
mixtures of Gaussians	90.48	97.01	75.56
symmetric KL	90.08	96.41	73.43

F.2. Additional ablation results *w.r.t.* hyperparameters

The effects of the hyperparameters λ , T , and γ on the final performance of CCDB are presented in Figure 11. We observe that: (i) As shown in Figure 11(a), $\lambda = 0$ consistently performs worse across all datasets, indicating that intra-class compactness is essential for improved performance. Values of $\lambda \in [0.1, 0.5]$ yield robust results. (ii) In Figure 11(b), performance drops drastically without clipping. This is because unbounded weights introduce excessive flexibility, leading to over-weighting of minority groups and degradation on majority groups. Empirically, binary tasks prefer $T \approx 1.5$, while 10-class tasks favor $T \approx 2.5$. (iii) As shown in Figure 11(c), $\gamma \in [0.3, 0.5]$ yields consistently robust performance across all datasets.

For hyperparameter selection, We respectfully clarify that our method does not rely on bias-annotated validation data. Worst-group accuracy (WGA) on test split is used only for analyzing hyperparameter sensitivity, not for model selection.


 Figure 11. The effect of λ .

In practice, for selecting hyperparameters such as T and γ , we use the biased predictions from the Stage 1 model on the validation set to partition samples into pseudo-groups based on their predicted and ground-truth labels, and monitor the WGA over these pseudo-groups to select the best setting. This procedure depends only on class labels and model predictions. For each hyperparameter setting, the best performance can be selected using either pseudo-WGA or worst-class accuracy, as discussed in Section D.

Trade-off between WGA and i.i.d. performance. Improvements in WGA achieved by CCDB may come at the cost of reduced i.i.d. accuracy. Such a trade-off is common in debiasing methods. In our framework, this trade-off can be controlled through the clipping threshold T or by interpolating between uniform and learned weights. The results on Waterbirds (Table 8) demonstrate a clear and controllable trade-off.

 Table 8. IID-WGA trade-off under different values of T on Waterbirds.

T	0	0.5	1.0	1.5
i.i.d.	97.88	93.98	91.97	92.59
WGA	73.41	85.10	89.22	90.48

F.3. CCDB with ground-truth reweighting

We consider an oracle baseline (termed CCDB-oracle) by replacing the estimated sample weights in CCDB with those obtained from ground-truth bias annotations. Results are reported in Table 9.

On Waterbirds and CelebA, CCDB-oracle serves as an upper bound of our method. On the other datasets, however, it underperforms CCDB, highlighting the importance of sample-level upweighting within each group.

Table 9. CCDB results with oracle(optimal) weights.

		cMNIST(5%)	cCIFAR10(5%)	Waterbirds	CelebA	CivilComments
IID	CCDB	96.37	74.64	92.59	90.08	83.60
	CCDB-oracle	86.76	60.87	93.82	93.58	84.96
WGA	CCDB	NA	NA	90.48	85.27	75.00
	CCDB-oracle	NA	NA	91.16	87.86	72.75

F.4. On freezing marginal parameters

We believe that freezing marginals functions as a regularization mechanism rather than imposing a biased prior. This design stabilizes optimization and prevents trivial or degenerate solutions, ensuring that alignment focuses on conditional structure

rather than drifting global distributions. While adaptive anchors (*e.g.*, reweighted marginals) are possible, they introduce additional instability due to feedback between weights and targets.

F.5. On Stage 1 fragility

Stage 1 is crucial for CCDB as it provides reliable pseudo-bias predictions. Our method is robust regardless of whether core features are easier to learn than spurious features or not. In particular, it does not suffer from the failure mode of mistakenly capturing core features and subsequently degrading them. As discussed in Figure 1 (right), our method remains robust under different outcomes of Stage 1: if spurious features are captured (Figure 1(a) and (c)), they provide useful signals for debiasing; if core features are captured (Figure 1(b)), the method reduces to a near-ERM solution without degrading performance, and moreover, the alignment objective emphasizes samples near the decision boundary when matching class-wise statistics, which helps improve performance in ambiguous regions rather than degrading the learned signal.

F.6. Further discussion on the limitations and potential negative social impact

While CCDB demonstrates strong empirical performance, several limitations and potential negative social impacts should be noted.

Our method relies on several practical design choices that introduce potential limitations. First, it depends on Stage 1 to provide a useful proxy for spurious factors; performance may degrade if this representation fails to capture relevant bias signals due to overemphasis on outliers or mislabeled samples (*i.e.* degrades to ERM). Second, we adopt a Gaussian moment-matching surrogate for tractability, which may be insufficient for highly multi-modal latent distributions. Third, the method involves hyperparameters $\{\gamma, T, \lambda\}$, which, although empirically stable with simple heuristics, may still require tuning depending on dataset characteristics.

In addition, we acknowledge that the method is not uniformly optimal across all benchmarks, and improvements in worst-group accuracy do not necessarily translate directly to fairness in real-world deployment, especially when minority patterns are noisy or underrepresented. This reflects the inherent trade-off between robustness and average accuracy, as well as the gap between benchmark robustness and real-world fairness. Therefore, caution should be exercised when applying the proposed method in high-stakes domains, where fairness and robustness requirements may differ substantially from the assumptions underlying standard benchmarks.